Ice-wedge polygon centres are more ordered than a random tessellation: structure-factor analysis of a million-polygon network with heterogeneity-limited long-range order

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Abstract

Ice-wedge polygons tile vast areas of permafrost lowland and are a textbook example of a self-organised fracture pattern with a characteristic length. How *ordered* that pattern is — in the precise sense used for disordered hyperuniform media — has not been quantified. Using a lidar-derived inventory of 1.16×10^6 ice-wedge polygons south of Prudhoe Bay, Alaska, we treat the polygon centres as a planar point pattern and compute the pair correlation $g(r)$, the structure factor $S(k)$ and the number variance, comparing against complete spatial randomness, an inhomogeneous-Poisson null matched to the local density, and — crucially — a Poisson–Voronoi tessellation (the centroids of a *random* tessellation of the same density). Polygon centres show hard-core exclusion and a characteristic spacing of $\sim$15 m, and a deep structure-factor suppression, $S_{\min} \approx 0.1$, at intermediate wavenumber. This suppression is genuine: it lies far below the inhomogeneous-Poisson null in every window analysed, and below the Poisson–Voronoi null in the most uniform windows — so the ice-wedge crack network is *more ordered than a random tessellation*. Long-range hyperuniformity is, however, *not* confirmed: $S(k \to 0)$ remains finite and the number variance grows faster than the window area ($\sigma^2 \propto R^{2.7}$), because real geomorphic heterogeneity — spatial variation in polygon size and coverage — dominates the largest scales; the suppression deepens in more uniform windows, indicating a heterogeneity-limited rather than intrinsic floor. Comparing intact (low-centred) and thermokarst-degraded (high-centred) terrain matched for uniformity, the intermediate-range order persists through degradation: thermokarst inverts the microtopographic relief but preserves the planimetric network order. Ice-wedge polygon centres are thus a locally rigid, intermediate-range-ordered disordered pattern with a hyperuniform tendency that is masked at long range by landscape heterogeneity.

1 Introduction

Where permafrost is ice-rich and the ground surface is flat, repeated winter thermal contraction cracks the frozen soil into a polygonal network; meltwater entering the cracks freezes, and over many years wedges of ice grow beneath the troughs, outlining *ice-wedge polygons* that cover large parts of the Arctic coastal plains (Lachenbruch, 1962). The pattern is self-organised and has a characteristic cell size set by the mechanics of thermal-contraction cracking and the relaxation of thermal stress (Lachenbruch, 1962; Plug & Werner, 2002); like other periglacial patterned ground, its *regular spacing* has long been recognised and modelled (Kessler & Werner, 2003; Plug & Werner, 2002).

Recognising a characteristic spacing, however, says little about how *ordered* the pattern is in the modern sense of point-pattern statistics. A pattern can possess a preferred neighbour distance yet remain essentially random at large scales (a liquid), or it can additionally suppress long-wavelength density fluctuations — the property of *hyperuniformity* (Torquato & Stillinger, 2003; Torquato, 2018), which is now recognised in systems from avian photoreceptors to dryland vegetation. Whether ice-wedge polygons, or any patterned ground, are hyperuniform — and whether their fracture network is more ordered than a *random* space-filling tessellation — has not, to our knowledge, been tested.

Here we treat the centres of 1.16×10^6 lidar-mapped ice-wedge polygons as a planar point process and quantify their order with the pair correlation, the structure factor and the number variance. The decisive comparison is against a Poisson–Voronoi tessellation: ice-wedge polygons *are* a space-filling tessellation, so the relevant question is not "are they more ordered than random points" (trivially yes) but "are they more ordered than a random *tessellation*". We also ask whether the order survives thermokarst degradation, using the relative elevation of each polygon centre as an intact-versus-degraded proxy.

2 Data

We use the lidar-derived ice-wedge polygon inventory of Abolt & Young (2020) (data at Abolt & Young, 2019), which maps $>10^6$ polygons over $\sim 1200\,\mathrm{km}^2$ of the Arctic Coastal Plain south of Prudhoe Bay, Alaska, from a 0.5 m digital elevation model using a convolutional neural network. The unified attribute table gives, for each of 1,162,700 polygons, the UTM (Zone 6N) coordinates of the centroid, the polygon area, and the relative elevation of the polygon centre with respect to its rim — a proxy for low-centred (intact, raised-rim) versus high-centred (thermokarst-degraded) form. The median polygon area is $182\,\mathrm{m}^2$ (equivalent diameter $\sim 15\,\mathrm{m}$), setting the characteristic spacing; 34% of polygons are low-centred and 66% high-centred (Fig. 1).

3 Methods

Window selection. The mapped polygons cover only ∼18% of the survey rectangle: they form continuous, space-filling patches separated by lakes, river corridors and non-polygonal terrain. These gaps impose large-scale density fluctuations that would dominate the structure factor at low wavenumber. We therefore restrict analysis to gap-free, internally homogeneous sub-windows, selecting 1.5 km windows by the coefficient of variation (CV) of polygon counts in 200 m cells (low CV = uniform coverage); we report results across the most uniform windows and, for the degradation analysis, across 90 non-overlapping windows.

Point-pattern statistics. Within each window we compute the isotropic pair correlation $g(r)$, the radially averaged structure factor $S(\mathbf{k}) = N^{-1}\big|\sum_j e^{-i\mathbf{k}\cdot\mathbf{r}_j}\big|^2$ evaluated at the allowed wavevectors of the window (which suppresses the window-shape artefact), and the number variance $\sigma^2(R)$ in disks of radius R. Hyperuniformity requires $S(k \to 0) \to 0$ and $\sigma^2(R)$ growing more slowly than R^2 (ideally as R^1 in two dimensions) (Torquato & Stillinger, 2003; Torquato, 2018).

Null models. We compare the observed statistics against four nulls at matched density: (i) complete spatial randomness (CSR, Poisson); (ii) an *inhomogeneous*-Poisson null generated by kernel-resampling the observed positions, which reproduces the window's large-scale density variation but

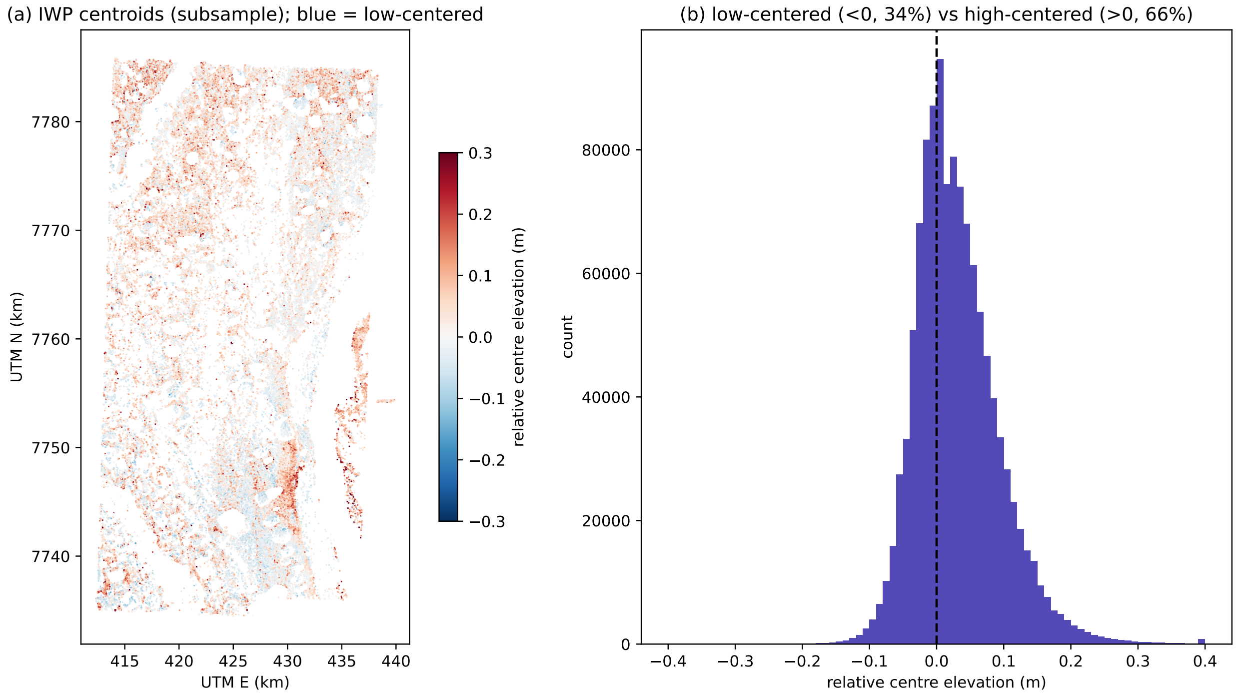

Figure 1: (a) Ice-wedge polygon centroids across the survey (random subsample), coloured by relative centre elevation (blue = low-centred/intact, red = high-centred/degraded). Polygons occur in continuous patches separated by lakes, drainage and non-polygonal ground. (b) Distribution of relative centre elevation.

no short-range order — a conservative control for heterogeneity; (iii) the centroids of a *Poisson–Voronoi* tessellation (Okabe et al., 2000), i.e. a random space-filling tessellation; and (iv) a hexagonal lattice (the ordered ceiling). The Poisson–Voronoi comparison is the key test of whether the ice-wedge network is more ordered than a random tessellation.

Degradation contrast. Because the intrinsic intermediate-range order ($S_{\min}$) is robust to large-scale heterogeneity whereas the pair-correlation peak height and the number-variance exponent are not (they correlate strongly with the density CV), we use $S_{\min}$ to compare intact and degraded terrain, matching low-centred and high-centred windows for CV.

4 Results

4.1 Local order and a characteristic spacing

Polygon centres show clear local order (Fig. 2a): the pair correlation has a hard core ($g = 0$ below $\sim$8 m), a sharp first peak at $\sim$15–17 m — the characteristic polygon spacing — and damped oscillations decaying to unity. This is the signature of a dense, exclusion-dominated liquid-like arrangement.

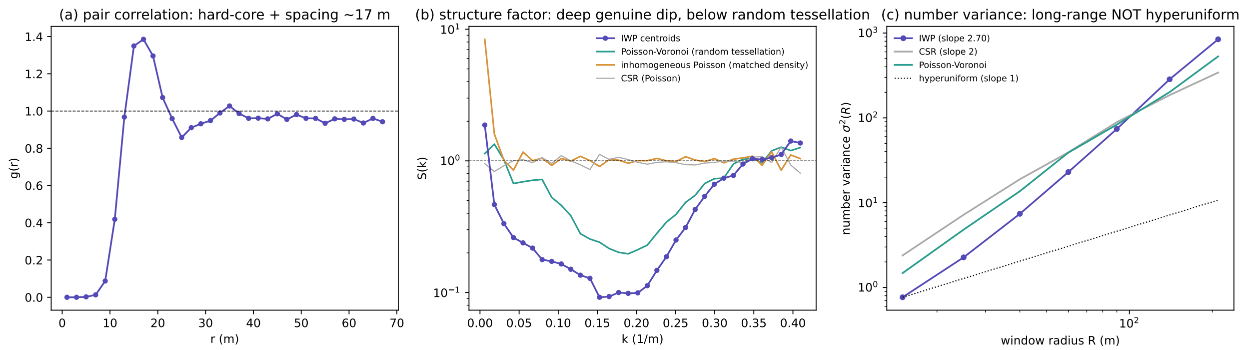

Figure 2: Order of ice-wedge polygon centres in a representative uniform window ($N = 7843$). (a) Pair correlation: hard-core exclusion and a characteristic spacing $\sim$17 m. (b) Structure factor: the observed $S(k)$ (purple) dips to $\approx$0.09 — far below the inhomogeneous-Poisson (orange) and CSR (grey) nulls, and below the Poisson–Voronoi null (teal) — so the network is more ordered than a random tessellation; $S(k)$ rises again as $k \to 0$. (c) Number variance grows as $R^{2.7}$, faster than the hyperuniform R^1 (dotted) and even than Poisson R^2 at large R: long-range order is not hyperuniform.

4.2 Intermediate-range suppression is genuine and exceeds a random tessellation

The structure factor (Fig. 2b) shows a deep suppression, with a minimum $S_{\min} \approx 0.09$–0.10 at $k \approx 0.16\,\mathrm{m}^{-1}$ (a scale of $\sim$40 m, about 2.5× the spacing) in the most uniform windows. Two controls establish that this is real order rather than an artefact of the window's density structure. First, $S_{\min}$ lies far below the inhomogeneous-Poisson null (which is ≈ 1 at the dip) in every window analysed: the suppression is not produced by the large-scale density variation. Second, in the most uniform windows $S_{\min}$ lies below the Poisson–Voronoi null ($\approx$0.10 versus $\approx$0.20): *the ice-wedge crack network is more ordered than a random space-filling tessellation of the same density.* The suppression is markedly stronger than that reported for some other geological spacing patterns (Sec. 5).

4.3 Long-range order is not hyperuniform

Despite the deep intermediate-range dip, the pattern is not hyperuniform at long range. The structure factor turns upward as $k \to 0$ rather than vanishing, and the number variance grows as $\sigma^2(R) \propto R^{2.7}$ (Fig. 2c) — faster than the hyperuniform R^1 and, at large R, faster even than the Poisson R^2, indicating super-Poissonian density fluctuations at the largest scales. These fluctuations reflect real geomorphic heterogeneity: polygon size and areal coverage vary across the landscape. Crucially, the suppression deepens systematically in more uniform windows, which shows that the long-range floor is set by heterogeneity (gaps, size variation), not by an intrinsic limit to the order. A clean test of hyperuniformity would require a region of genuinely uniform polygon size, which this mature, partly degraded landscape does not provide.

4.4 The order survives thermokarst degradation

Comparing intact (low-centred) and degraded (high-centred) terrain matched for density uniformity (Fig. 3), both show the same hard-core spacing and a clear intermediate-range structure-factor dip; the difference in $S_{\min}$ is weak and not significant (medians 0.29 versus 0.22, with strongly overlapping distributions). The order is therefore not destroyed by thermokarst: as polygons evolve from low-

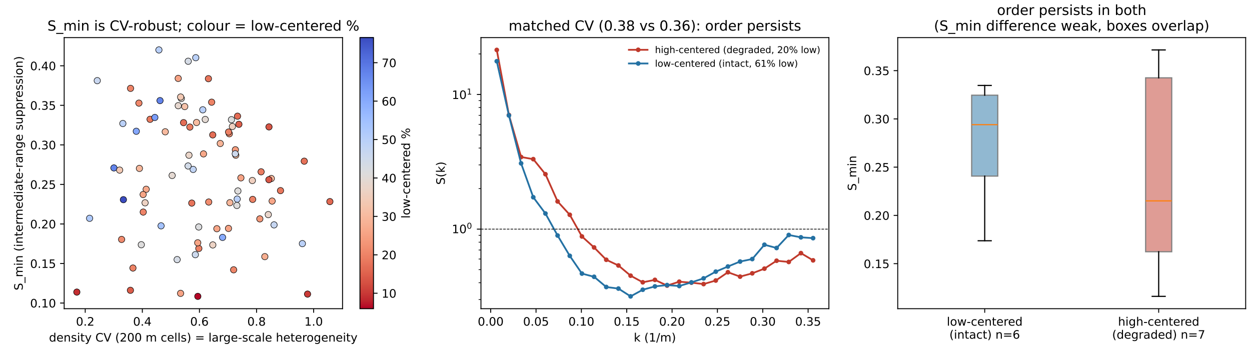

Figure 3: Degradation contrast. (a) $S_{\min}$ versus the density CV across 90 windows, coloured by low-centred fraction: $S_{\min}$ is insensitive to CV (a clean intrinsic-order metric) and shows no strong dependence on degradation state. (b) Structure factor for an intact (low-centred) and a degraded (high-centred) window matched in CV: both dip clearly — the order persists. (c) $S_{\min}$ for intact versus degraded windows in a matched-CV band: the distributions overlap, so degradation does not destroy the intermediate-range order.

to high-centred, the microtopographic relief inverts but the planimetric ice-wedge network — and hence the spacing order of the centres — persists, consistent with the troughs deepening rather than disappearing during degradation.

5 Discussion

A fracture network more ordered than random. The central result is that ice-wedge polygon centres are more ordered than a Poisson–Voronoi tessellation. This is the meaningful comparison: exceeding complete spatial randomness would be unremarkable, because the centroids of *any* space-filling tessellation avoid one another and so appear sub-Poissonian; a random tessellation (Poisson–Voronoi) already carries that built-in short-range order, so beating it requires the crack network itself to be regularised. Thermal-contraction cracking is a sequential, stress-relaxing process: each new crack preferentially bisects the largest uncracked domains and tends to meet existing cracks orthogonally, producing cells more uniform in size than those of a random tessellation (Lachenbruch, 1962; Plug & Werner, 2002). The hard core in $g(r)$ is the geometric expression of this mechanism — a thermal stress-relief halo: once a primary contraction crack forms it relaxes the tensile stress in its surroundings, so an adjacent parallel crack cannot nucleate within a characteristic distance (Lachenbruch, 1962). The deep, genuine $S(k)$ suppression ($S_{\min} \approx 0.1$) is the quantitative signature of that regularisation. We read this as a network that is geometrically optimised for strain relief, while noting that the static spatial statistics measured here constrain the *outcome* of that process, not its dynamics directly.

Hyperuniform tendency, heterogeneity-limited. The pattern has a strong hyperuniform *tendency* at intermediate scales but does not reach confirmed long-range hyperuniformity: $S(k \to 0)$ stays finite and the number variance is super-linear, because landscape-scale variation in polygon size and coverage injects long-wavelength density fluctuations. That the suppression deepens as windows become more uniform identifies this as an extrinsic, heterogeneity-imposed floor rather than an intrinsic property of the cracking process. Rather than a mere shortfall from a mathematical ideal, this long-range heterogeneity is a geomorphic fingerprint: the wavenumber at which $S(k)$ turns up from its minimum sets a length scale for the homogeneous polygonal patches, which on the

Arctic Coastal Plain are bounded by drained thaw-lake basins, drainage networks and gradients in ground-ice content. The local-versus-long-range distinction — robust local and intermediate-range order, long-range order limited by real heterogeneity — recurs across natural two-dimensional spacing patterns and should be kept explicit when hyperuniformity is claimed for field data.

Comparison with other geological spacing patterns. Among geological map patterns analysed with the same tools, ice-wedge polygons are the most strongly suppressed at intermediate range. Salt structures, spaced by a buoyancy (Rayleigh–Taylor) instability, form an isotropic hard-core, liquid-like regular pattern whose intermediate-range suppression is shallower ($S_{\min} \approx 0.4$; Chen, 2026a); drumlins, shaped by directional subglacial shear, are anisotropic and only steric-regular across flow (Chen, 2026b). Ice-wedge polygons, an isotropic fracture tessellation with no imposed flow direction and no intrinsic basin-scale density gradient, achieve the deepest suppression ($S_{\min} \approx 0.1$). None of these field patterns, however, reaches *confirmed* long-range hyperuniformity: in every case real geomorphic heterogeneity sets the long-wavelength floor.

6 Conclusions

- Ice-wedge polygon centres show hard-core exclusion and a characteristic spacing of $\sim$15 m, with a deep, genuine structure-factor suppression ($S_{\min} \approx 0.1$) at intermediate wavenumber.
- In uniform windows the suppression lies below the Poisson–Voronoi null: the ice-wedge crack network is *more ordered than a random space-filling tessellation.*
- Long-range hyperuniformity is not confirmed — $S(k \to 0)$ is finite and $\sigma^2(R) \propto R^{2.7}$ — because landscape heterogeneity dominates the largest scales; the suppression deepens in more uniform windows, marking a heterogeneity-limited floor.
- The intermediate-range order persists through the low-to-high-centred thermokarst transition: degradation inverts the relief but preserves the planimetric network order.

Data availability

The ice-wedge polygon inventory is from Abolt & Young (2020), with data archived at PANGAEA (Abolt & Young, 2019) under CC-BY-4.0. All analysis code and derived per-window statistics are openly available at `https://github.com/Ruqing1963/ice-wedge-polygon-order`. Analysis used only open scientific Python (NumPy, SciPy, Matplotlib).

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